Technical Assessment - StreamAMG

Overview

This technical assessment evaluates your engineering approach, technical skills, and how you leverage AI tooling in your development workflow. Please don't spend more than 2 hours on this - we want to see your approach, not a polished production system. If you choose to spend longer, that's your call.

Our Tech Stack

At StreamAMG, we typically work with:

- TypeScript/Angular, React and React Native
- AWS with infrastructure-as-code
- Containerised and serverless patterns
- AI development tools (Claude Code, Cursor, GitHub Copilot, etc.)

You're welcome to use this stack or demonstrate your strongest work in an alternative. If you choose something different, briefly explain why.

Challenge: Simple Streaming Asset & Playback API

no frontend...

Background

StreamAMG builds media services and streaming products. Traffic patterns can be spiky (e.g., live events, kick-offs, peak viewing windows) and systems need to be reliable, observable and secure.

For this assessment, you'll build a simplified proof-of-concept API that demonstrates how you approach a streaming-adjacent problem.

NodeJS + ExpressJS (TS)

Requirements

"Creating" = POST /assets
"Retrieving" = GET /assets
GET /assets/:id

Create a simplified RESTful API that manages media content data:

- Basic CRUD operations for media assets (focus on creating and retrieving only)
- Simple filtering capability for at least one type (e.g., by asset status or content type)
- A basic event emitter for asset state changes (e.g., processing complete, publish/unpublish)

/api/assets?status=
/api/assets?contentType=

log/notification

EventEmitter

Implementation

Build a working solution that demonstrates:

folders, separation of concerns, readable

- Clean, well-structured code
- Appropriate test coverage

~ 90-95%
E2E Jest Supertest

- Clear API documentation (endpoints and example requests/responses)

OpenAPI Spec

- An event-driven component (e.g., asset updates triggering downstream actions)

simple map

Use an in-memory database for simplicity. Beyond that, make your own decisions about structure, patterns, and how far to take the implementation - we're interested in seeing your judgment.

Architecture

- Provide a brief explanation of your architecture choices (1-2 paragraphs)
- Keep your solution simple and focused on demonstrating core concepts
- In your README, briefly describe (don't implement) how your solution could scale during peak usage periods

AI Usage

We expect you to use AI tools throughout this assessment. This reflects how we work at StreamAMG.

What we want to see:

- Which AI tools you used (Claude Code, Cursor, Copilot, ChatGPT, etc.)
- Key prompts or conversations that shaped your solution - include these as artefacts (screenshots, exported chats, or a summary document)
- How you iterated on AI-generated code - what did you accept, reject, or refine?
- Where AI helped most and where you needed to take over

Being thoughtful about AI usage is part of the assessment, not a side note.

Assessment Criteria

We evaluate your submission on:

- **Code quality**: Clean, maintainable code following best practices
- **Architecture**: Appropriate design choices for the problem
- **Testing**: Meaningful test coverage demonstrating your approach to quality
- **Documentation**: Clear explanation of your solution

- **AI-assisted workflow**: Effective use of AI tools with clear artefacts showing your process - how you direct AI, validate output, and iterate
- **Problem-solving**: How you approach the challenge and the judgment calls you make

We're assessing the same criteria for both all our Engineering roles. We expect Senior and Engineering Manager candidates to demonstrate greater depth in architectural thinking, code quality, and sophistication in how they leverage AI tooling.

Submission Instructions

Create a GitHub repository containing:

- Your solution code
- A README with:
  - Setup and run instructions
  - Brief architecture explanation (1-2 paragraphs)
  - Key technical decisions (bullet points)
  - AI artefacts and workflow documentation
  - Assumptions made
- Ability to test, build, and run locally

Submit the repository link by the deadline provided.

Time Expectations

We respect your time - please don't spend more than 2 hours on this. Focus on demonstrating your approach and AI workflow rather than implementing every feature. Quality over quantity.

Questions

If you have any questions or need clarification, please contact:

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Good luck - we look forward to seeing your solution!